

Year 7 · Chapter 3 · Whole numbers: answers (teacher copy)

3.01 Rounding and estimation (booklet p. 3)

WE DO examples

- Ex 1: 88 000
- Ex 2: 87 530
- Ex 3: $400 + 80 + 900 + 20 = 1400$
- Ex 4: $900 + 300 - 400 = 800$
- Ex 5: $90 \times 20 = 1800$
- Ex 6: $2400 \div 30 = 80$

Set A

- 50
- 150
- 3 880
- 1 000
- 6 000
- 87 530

Set B

- 700
- 87 500
- 3 000
- 90 000
- 65 000
- 150 000

Set C

- $600 + 300 = 900$
- $4000 - 2000 = 2000$
- $40 + 70 + 400 = 510$
- $800 - 200 + 100 = 700$
- $7000 + 3000 = 10\,000$
- $5000 - 4000 = 1000$

Set D

- $50 \times 20 = 1000$ (exact 1008)
- $600 \times 9 = 5400$ (exact 5508)
- $4000 \div 40 = 100$ (exact ≈ 103.9)
- $30 \times 30 = 900$ (exact 868)

Extension

- 11 500 and 12 499
- 35 499

Exit ticket

- Level 1: 6400
- Level 2: $500 + 300 - 100 = 700$
- Level 3: $4000 \div 20 = 200$

3.02 Multiplication (booklet p. 7)

WE DO examples

- Ex 1: 48
- Ex 2: 200
- Ex 3: 170
- Ex 4: 297
- Ex 5: 882
- Ex 6: 10 170

Set A

- 48
- 90
- 180
- 88
- 100
- 350

Set B

- 170
- 200
- 297
- 80
- 180
- 231

Set C

- 882
- 832
- 2 250
- 744
- 3 702
- 4 131

Set D

- 10 170
- 1 081
- 14 040
- 1 368 people

Extension

- 3 ($37 \times 5 = 185$)
- e.g. 25×40 or 20×50

Exit ticket

- Level 1: 70
- Level 2: 1 458
- Level 3: 3 456

3.03 Division (booklet p. 11)

WE DO examples

- Ex 1: 240
- Ex 2: 16
- Ex 3: 124
- Ex 4: 132
- Ex 5: 162
- Ex 6: 121

Set A

- 240
- 8
- 9
- 35
- 12
- 7

Set B

- 16
- 18
- 30
- 32
- 81
- 12

Set C

- 124
- 132
- 162
- 121
- 235
- 156

Set D

1. 79 r 1
2. 250 r 3
3. 714 r 2
4. 6 buses ($250 \div 45 = 5 \text{ r } 25$)

Extension

1. 26 or 38
2. 6

Exit ticket

1. Level 1: 36
2. Level 2: 123
3. Level 3: 156 r 2

3.04 Divisibility tests (booklet p. 15)

WE DO examples

1. Ex 1: yes; no; no
2. Ex 2: no (digit sum 11); no
3. Ex 3: yes (08); yes ($208 \div 8 = 26$)
4. Ex 4: no (not divisible by 3)
5. Ex 5: B: 24
6. Ex 6: 2 (522)

Set A

1. yes
2. no
3. yes
4. yes
5. no
6. yes

Set B

1. yes (12)
2. no
3. yes (18)
4. no (1)
5. yes (12)
6. yes (18)

Set C

1. by 4, 6
2. by 4, 8
3. by 4, 6
4. by 4, 8
5. by 6
6. by 4, 8

Set D

1. by 2, 3, 4, 5, 6, 8, 9, 10
2. 24, 96, 120
3. 1, 4, 7
4. No: $9 = 3 \times 3$ so anything divisible by 9 is divisible by 3

Extension

1. Yes: ends in 0 (by 5) and digit sum 18 (by 3)
2. 1020

Exit ticket

1. Level 1: yes
2. Level 2: yes (digit sum 18)
3. Level 3: yes: even and digit sum 15

3.05 Long division (booklet p. 19)

WE DO examples

1. Ex 1: 26
2. Ex 2: 124
3. Ex 3: 256 r 7
4. Ex 4: 54
5. Ex 5: 156 r 8
6. Ex 6: 23 buses

Set A

1. 84
2. 150
3. 96
4. 120
5. 96
6. 144

Set B

1. 8
2. 7
3. 6
4. 16
5. 8
6. 9

Set C

1. 26
2. 124
3. 54
4. 112
5. 255
6. 234

Set D

1. 256 r 7
2. 156 r 8
3. 470 r 6
4. 23 buses

Extension

1. 994
2. 993

Exit ticket

1. Level 1: 6
2. Level 2: 108
3. Level 3: 148 r 4

3.06 Roman numerals (booklet p. 23)

WE DO examples

1. Ex 1: 37
2. Ex 2: 240
3. Ex 3: 24
4. Ex 4: XXVIII
5. Ex 5: LXXIV
6. Ex 6: CXXV

Set A

1. 8
2. 12
3. 37
4. 60
5. 210
6. 15

Set B

1. 4
2. 9
3. 24
4. 40
5. 240
6. 90

Set C

1. VI
2. XIII
3. XXVIII
4. LXXIV
5. CXXV
6. XLIX

Set D

1. MCMXCIX
2. 2026
3. e.g. 2026 = MMXXVI
4. Only I before V or X is allowed (99 = XCIX)

Extension

1. CCLV
2. CVIII

Exit ticket

1. Level 1: 16
2. Level 2: XXXIX
3. Level 3: MMXXIV

3.07 Index notation and powers (booklet p. 27)**WE DO examples**

1. Ex 1: 3^4
2. Ex 2: 7^7
3. Ex 3: 36
4. Ex 4: 125
5. Ex 5: 100 000
6. Ex 6: 256

Set A

1. 4^3
2. 6^5
3. 9^2
4. 2^6
5. 10^4
6. $5^3 \times 3^2$

Set B

1. $8 \times 8 \times 8$
2. $3 \times 3 \times 3 \times 3 \times 3$
3. 11×11
4. $2 \times 2 \times 2 \times 2$
5. 7
6. $2 \times 2 \times 2 \times 5 \times 5$

Set C

1. 36
2. 125
3. 32
4. 10 000
5. 81
6. 1

Set D

1. 256
2. $64 - 27 = 37$
3. $8 \times 9 = 72$
4. $2^5 = 32$ is larger than $5^2 = 25$

Extension

1. $3^4 = 9^2$
2. $n = 10$

Exit ticket

1. Level 1: 7^3
2. Level 2: 64
3. Level 3: $9 \times 16 = 144$

3.08 Square roots and cube roots (booklet p. 31)**WE DO examples**

1. Ex 1: 7
2. Ex 2: 12
3. Ex 3: 20
4. Ex 4: 9
5. Ex 5: $\sqrt{144} = 12$
6. Ex 6: 7 and 8

Set A

1. 4
2. 7
3. 9
4. 10
5. 12
6. 1

Set B

1. 2
2. 3
3. 4
4. 5
5. 10
6. 1

Set C

1. 20
2. 20
3. 9
4. 15
5. $6 + 8 = 14$
6. $6 \times 2 = 12$

Set D

1. 9 and 10
2. 4 and 5
3. $\sqrt{100} = 10$; no $(6 + 8 = 14)$
4. 14 cm

Extension

1. 70 and 30
2. 7 cm

Exit ticket

1. Level 1: 8
2. Level 2: 5
3. Level 3: $11 - 10 = 1$

3.09 Prime and composite numbers (booklet

p. 35)

WE DO examples

1. Ex 1: 1, 2, 3, 4, 6, 12; 1, 13
2. Ex 2: neither
3. Ex 3: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
4. Ex 4: No: $51 = 3 \times 17$
5. Ex 5: 29, 31, 37, 41, 43, 47
6. Ex 6: 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58

Set A

1. 1, 2, 3, 6
2. 1, 11
3. 1, 3, 5, 15
4. 1, 2, 4, 5, 10, 20
5. 1, 17
6. 1, 2, 3, 4, 6, 9, 12, 18, 36

Set B

1. prime
2. composite
3. prime
4. composite
5. prime
6. composite

Set C

1. 53, 59, 61, 67
2. 21, 22, 24, 25, 26, 27, 28, 30
3. 42, 44, 45, 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58
4. 49, 51, 55, 57, 63, 65, 69

Set D

1. every other even number has 2 as a factor
2. e.g. $5 + 31$, $7 + 29$, $13 + 23$, $17 + 19$
3. 3,5; 5,7; 11,13; 17,19; 29,31; 41,43
4. yes, 31

Extension

1. yes (not divisible by 2, 3, 5, 7, 11, 13)
2. no: 13×17

Exit ticket

1. Level 1: prime
2. Level 2: 1, 2, 4, 7, 14, 28
3. Level 3: 31, 37, 41, 43

3.10 Prime factors (booklet p. 39)

WE DO examples

1. Ex 1: $2^2 \times 3 \times 5$
2. Ex 2: $2^3 \times 5^2$
3. Ex 3: $2^4 \times 3^2$
4. Ex 4: $2 \times 3^2 \times 5$
5. Ex 5: 2^8 , so $2^4 = 16$
6. Ex 6: $2^3 \times 3^6$, so $2 \times 3^2 = 18$

Set A

1. $2^2 \times 3$
2. 2×3^2
3. $2 \times 3 \times 5$
4. $2^3 \times 5$
5. $3^2 \times 5$
6. 2×5^2

Set B

1. $2^3 \times 3$
2. $2^2 \times 3^2$
3. $2^3 \times 3^2$

Set C

1. $2^2 \times 3 \times 5$
2. $2^3 \times 5^2$
3. $2^4 \times 3^2$
4. $2^2 \times 3^2 \times 7$
5. $3^3 \times 5^2$
6. $2^3 \times 7^2$

Set D

1. $2^8 \rightarrow 16$
2. $2^2 \times 3^4 \rightarrow 18$
3. $2^3 \times 3^6 \rightarrow 18$
4. $2^3 \times 7^3 \rightarrow 14$

Extension

1. 5
2. 12

Exit ticket

1. Level 1: $2^2 \times 5$
2. Level 2: $2^2 \times 3 \times 7$
3. Level 3: $2^6 \times 3^2 \rightarrow 24$

3.11 Highest common factor (booklet p. 43)

WE DO examples

1. Ex 1: 6
2. Ex 2: 4
3. Ex 3: 12
4. Ex 4: 17
5. Ex 5: 81
6. Ex 6: 36

Set A

1. 3
2. 4
3. 5
4. 7
5. 8
6. 1

Set B

1. 4
2. 12
3. 15
4. 8
5. 18
6. 14

Set C

1. 81
2. 40
3. 48
4. 42
5. 75
6. 14

Set D

1. $\frac{3}{7}$
2. 8
3. 15 cm
4. e.g. 24 and 30, or 30 and 36

Extension

1. 24
2. 15

Exit ticket

1. Level 1: 4
2. Level 2: 14
3. Level 3: 84

3.12 Lowest common multiple (booklet p. 47)**WE DO examples**

1. Ex 1: 12
2. Ex 2: 24
3. Ex 3: 90
4. Ex 4: 35
5. Ex 5: 400
6. Ex 6: 180

Set A

1. 3, 6, 9, 12, 15, 18
2. 5, 10, 15, 20, 25, 30
3. 8, 16, 24, 32, 40, 48
4. 12, 24, 36, 48, 60, 72
5. 9, 18, 27, 36, 45, 54
6. 15, 30, 45, 60, 75, 90

Set B

1. 6
2. 20
3. 18
4. 21
5. 24
6. 30

Set C

1. 90
2. 400
3. 180
4. 42
5. 72
6. 180

Set D

1. 7:00 am (60 minutes)
2. 24
3. 4 packs of hot dogs, 5 packs of buns (40 each)
4. 3, 6, 12, 24

Extension

1. 72
2. 60 seconds

Exit ticket

1. Level 1: 12
2. Level 2: 48
3. Level 3: 84